

Reviewing the Project

When you're beginning a new phase of development on a large project, it's always a good idea to revisit your plan and clarify what you want to accomplish with the code you're about to write. In this chapter, we'll do the following:

- Add a single alien to the top-left corner of the screen, with appropriate spacing around it.
- Fill the upper portion of the screen with as many aliens as we can fit horizontally. We'll then create additional rows of aliens until we have a full fleet.
- Make the fleet move sideways and down until the entire fleet is shot down, an alien hits the ship, or an alien reaches the ground. If the entire fleet is shot down, we'll create a new fleet. If an alien hits the ship or the ground, we'll destroy the ship and create a new fleet.
- Limit the number of ships the player can use, and end the game when the player has used up the allotted number of ships.

We'll refine this plan as we implement features, but this is specific enough to start writing code.

You should also review your existing code when you begin working on a new series of features in a project. Because each new phase typically makes a project more complex, it's best to clean up any cluttered or inefficient code. We've been refactoring as we go, so there isn't any code that we need to refactor at this point.

Creating the First Alien

Placing one alien on the screen is like placing a ship on the screen. Each alien's behavior is controlled by a class called `Alien`, which we'll structure like the `Ship` class. We'll continue using bitmap images for simplicity. You can find your own image for an alien or use the one shown in Figure 13-1, which is available in the book's resources at https://ehmatthes.github.io/pcc_3e. This image has a gray background, which matches the screen's background color. Make sure you save the image file you choose in the `images` folder.



Figure 13-1: The alien we'll use to build the fleet